

"Vedanta is committed to making India self-reliant in electronics. This is the beginning of the creation of a Silicon Valley in India, a cutting-edge and world-class electronics ecosystem"

– ANIL AGARWAL

Chairman, Vedanta, while announcing its semiconductors and display fab manufacturing venture in July 2023

conductors in the country is estimated to nearly triple from $22.7 billion (Rs 1.9 lakh crore) in 2019 to $64 billion (Rs 5.3 lakh crore) by 2026.

"Around the world, countries have realised that semiconductors are vital for economic and national security. How chips are made is as important as where chips are made," was how Dr Prabhu Raja, president of the Semiconductor Products Group at the US-based firm Applied Materials, put it at the Semicon India 2023 conference in July.

₹5.3 LAKH CROORE

Projected size of India's semiconductor market by 2026

₹1.3 LAKH CROORE

Worth of semiconductor chips imported in 2022-23

₹76,000 CRORE

Outlay for semiconductor PLI scheme announced in Dec. 2021

WHAT INDIA NEEDS TO MASTER

The semiconductor ecosystem comprises three areas: design, wafer manufacturing and semiconductor testing and packaging. Chip designing is where India currently has an advantage—over the past 20 years, several of the top global semiconductor design companies have set up centres in India. "We have 20-25 per cent of the world's chip designers in India," says Sanjay Gupta, chairman of the India Electronics and Semiconductor Association (IESA). The number of silicon designers with core expertise is expected to be 100,000-150,000, he adds. The idea, therefore, is to promote homegrown 'fabless' companies that can design chips by tapping into this rich talent pool in the domestic ecosystem. But ensuring access to capital for such start-ups will be a key factor.

Next is manufacturing, where lies the big challenge, something the Centre's PLI scheme hopes to address. Besides huge capital, setting up a chip foundry requires sophisticated technology. "Imagine you are packing millions of independent structures on a stamp-sized chip. This is done by very sophisticated robotics-driven machines," explains Gupta. The good news, he says, is that global firms with expertise in making these tools currently have a footprint in India and plan to expand their presence here. Besides chips, a related manufacturing subsidy scheme is for the fabrication of thin-film transistor or liquid crystal display screens used in phones, laptops and other devices. These, too, are currently imported.

THE CHANGE-MAKERS

VEDANTA

The Indian multinational in July announced its venture into semiconductors and display fab manufacturing via special-purpose vehicles under a group firm, Twin Star Technologies Ltd; has signed an MoU with Gujarat to set up a facility in Dholera

HCL

The Shiv Nadar-led IT services and consulting group is also venuring into the semiconductor manufacturing space; is among four companies to have registered for a government incentive scheme for semiconductor packaging

MICRON TECHNOLOGY

The US-based chip manufacturer plans to invest $825 million (Rs 6,845 crore) to set up a semiconductor assembly and test facility in Gujarat with the first phase commencing operations by late 2024

The third vertical is Assembly, Testing, Marking and Packaging (ATMP), which refers to the process of adding various components to silicon wafers before they are ready for use on motherboards. In all, four schemes under India's semiconductor programme cater to all these segments. Chip-making, however, is an area of long gestation and payback, and demands large and sustained investments. For India, the key task is to build an ecosystem ground up. ■